

Course 1: Generalities about AI



IMT Atlantique
Bretagne-Pays de la Loire
École Mines-Télécom

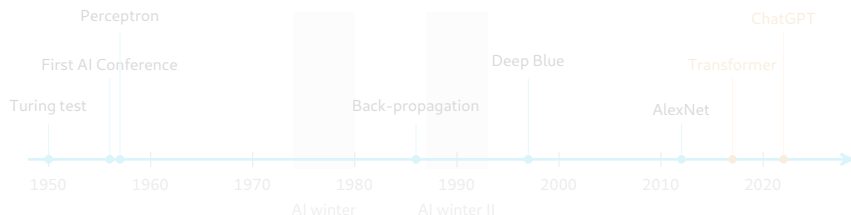
Artificial Intelligence?

A tentative definition of intelligence

- **Intelligence:** ability to **extract knowledge** from observations
- This knowledge is used to **solve tasks in different contexts and environments** (automation)

So AI: systems that solve tasks automatically. This definition is **very broad!**

AI is not a recent field!



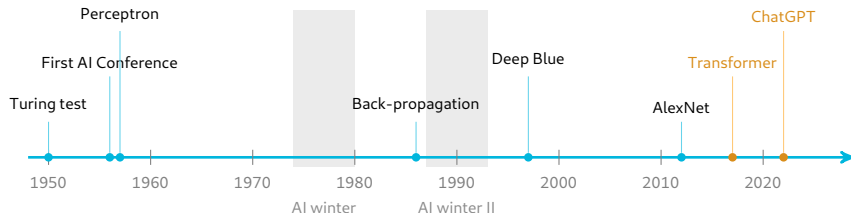
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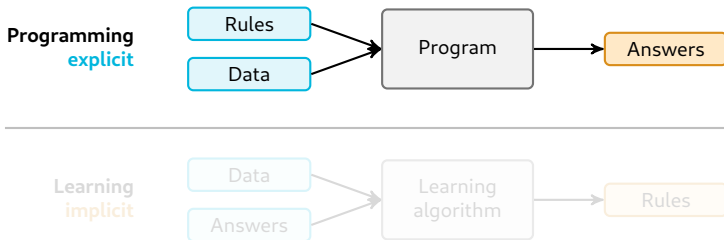
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Two ways of automation



A human writes the rules

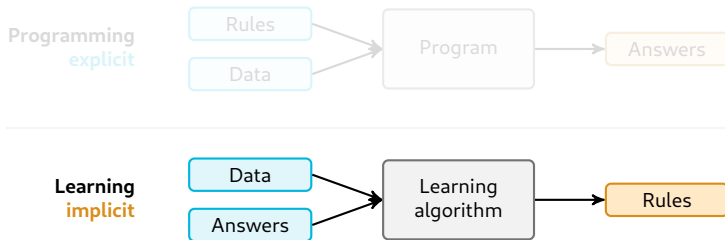
- You know exactly what happens (exact, verifiable, debuggable)
- You must be able to **state** the rules

The machine infers the rules

- Works when nobody can state them
- Needs data, and the rules stay implicit (hard to audit)

AI is generally understood as the second way: learning from data.

Two ways of automation



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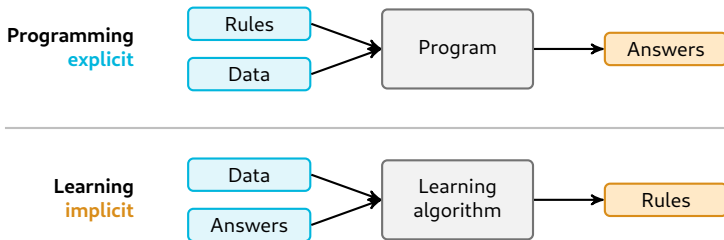
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"Learning", defined formally

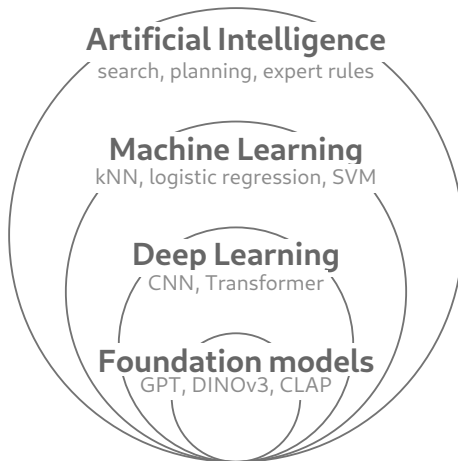
A program **learns** from experience E with respect to a task T and a performance measure P , if its performance at T , measured by P , **improves with** E . (Tom Mitchell, 1997)

T : the **task**
what you predict

E : the **data**
what you learn from

P : the **metric**
how you are judged

AI, Machine Learning & Deep Learning



Adapted from *Deep Learning*, 2016 by A. Courville, I. Goodfellow and Y. Bengio

Traditional application domains of AI

Vision

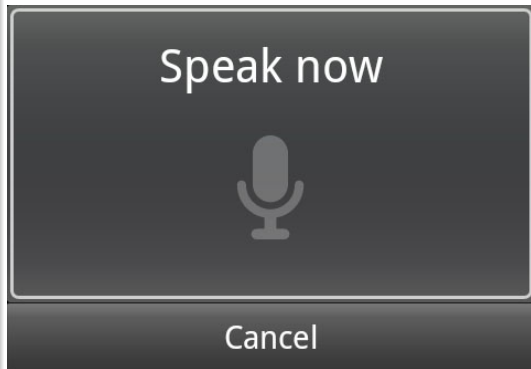
- Semantic Segmentation,
- Object/face recognition,
- Autonomous vehicles,
- Automatic diagnostic,
- Defects identification...



Traditional application domains of AI

Natural Language Processing (NLP)

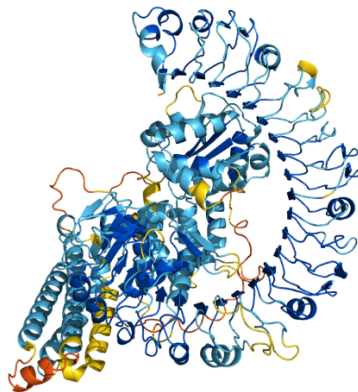
- Automatic assistant (agents),
- Voice-to-text,
- Automatic translation,
- Automatic summarizing,
- Sentiment analysis,
- Text indexing...



Traditional application domains of AI

Tons of other domains...

- Medical imaging,
- Games (chess, go, video games...),
- Audio and Music processing,
- Bioinformatics (AlphaFold),
- Recommender systems,
- Market analysis...



<https://alphafold.ebi.ac.uk/>

Generative AI: a recent breakthrough

Text Synthesis with Large Language Models (LLMs)

By now, you should have all heard about LLMs: ChatGPT (OpenAI), Claude (Anthropic), Gemini (Google), LLaMA (Meta), Mistral 8x7B (MistralAI), ... And many others!



Write an introduction to a master course on artificial intelligence for an engineering school



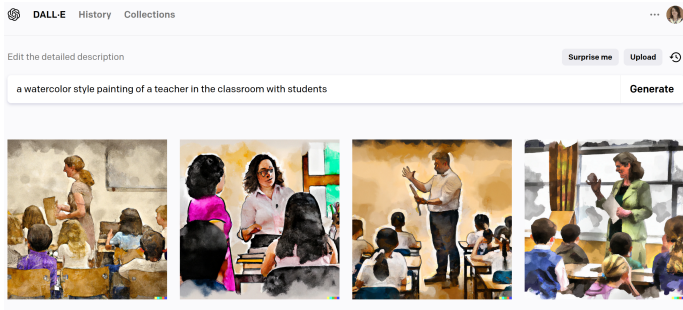
Welcome to the Master's course on Artificial Intelligence (AI) at our engineering school. In this program, you will gain a comprehensive understanding of the latest advances in the field of AI and machine learning. You will learn to design and implement intelligent systems using a variety of techniques and algorithms, and you will learn to critically evaluate the performance and limitations of these systems. The course will cover a wide range of topics including supervised and unsupervised learning, deep learning, natural language processing, computer vision, and robotics. You will also have the opportunity to work on real-world projects and to collaborate with other students and faculty members on cutting-edge research. By the end of the program, you will have the skills and knowledge needed to excel in a career in AI or to pursue further research in the field.

Generative AI: a recent breakthrough

Image Synthesis

- Stable Diffusion Models (<https://arxiv.org/pdf/2403.03206>)
- DALL-E (<https://labs.openai.com/>)
- Midjourney (<https://docs.midjourney.com/>)

Possible thanks to **multimodality**: manipulating simultaneously both text and image!



Generative AI: a recent breakthrough

Many other modalities...

- Speech (Many actors, ex: ElevenLabs, Kyutai)
- Music (StableAudio, Udio, Suno, ...)
- Video (StableVideo, Gemini Omni, ...)
- ...
- **Multimodal** models are arising

Some key open challenges (core AI research)

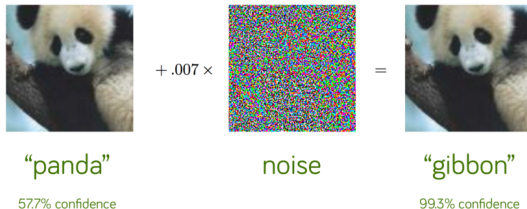
Interpretability



A trained model might be very accurate, but how does it take its decision ?

Some key open challenges (core AI research)

Learning what should be learned (robustness / adversarial attacks)



Random noise added to input images can dramatically change the result.

"Intriguing properties of neural networks", Arxiv research report, 2013.

Some key open challenges (core AI research)

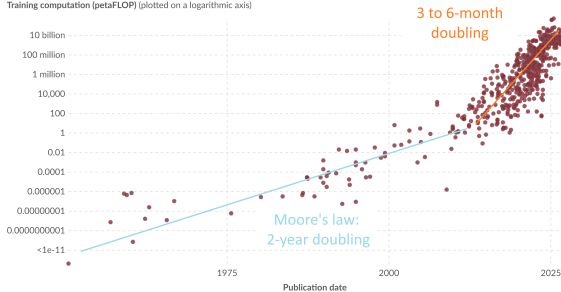
Computational and memory footprints

Computation used to train notable artificial intelligence systems

Our World
in Data

Computation is measured in total petaFLOP, which is 10^{15} floating-point operations¹. Estimated from AI literature, albeit with some uncertainty. Estimates are expected to be accurate within a factor of 2, or a factor of 5 for recent undisclosed models like GPT-4.

Training computation (petaFLOP) (plotted on a logarithmic axis)



Data source: Epoch AI (2026)

OurWorldinData.org/artificial-intelligence | CC BY

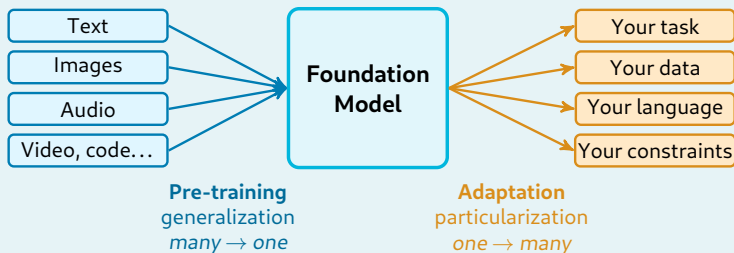
¹ Floating-point operation A floating-point operation (FLOP) is a type of computer operation. One FLOP represents a single arithmetic operation involving floating-point numbers, such as addition, subtraction, multiplication, or division.

Training a large algorithm: thousands to millions of parameters using Gigabytes of data.

Some key open challenges (core AI research)

Yesterday's challenge: make **one** model *generalize* beyond its training set.

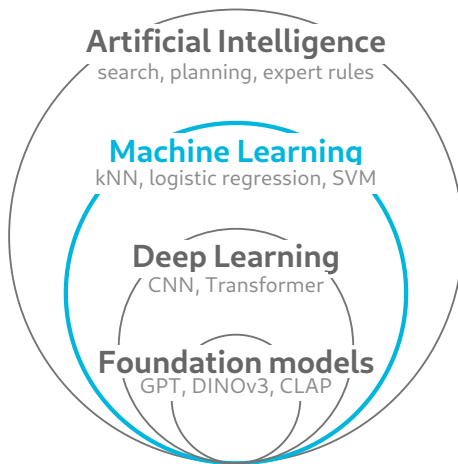
From **generalization** to **particularization**



Today: generalization largely comes with scale — the challenge is to *particularize* a generic model towards **your** problem, with little data and little compute.

How? Prompting, transfer learning, fine-tuning, PEFT/LoRA, RAG... (see course 5).

Let's dive into details!



Machine Learning

Examples

- Learning to play chess through playing games,
- Learning to recognize dogs and cats in images from annotated examples ...

Machine learning

- **Supervised:** Learning from **inputs** and **human annotations**
- **Unsupervised:** Learning from **inputs** only (patterns)
- **Self-supervised:** Learning by reconstructing the **inputs** from distorted versions

Generalization

- Generalization refers to the ability to infer **good decisions or representations** from **examples**.
- Goal: transfer knowledge from a dataset to another

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Global formalism

Input/output

- **Goal:** infer a function of parameters $\mathbf{W}$ from an input (often tensor) space to an output (often tensor) space, $\hat{\mathbf{y}} = f(\mathbf{x}, \mathbf{W})$.
- **Example:** input can be an image, output a vector where the largest value indicate the category the image belongs to.

Error/Loss

- **Loss $\mathcal{L}$:** nonnegative measure of the discrepancy between expected output $\mathbf{y}$ and obtained output $\hat{\mathbf{y}}$.
- **Example:** $\mathbf{y} = [0, 1]$ but $\hat{\mathbf{y}} = [0.2, 0.8]$.

Parameters

- $f(\cdot, \mathbf{W})$ contains **parameters $\mathbf{W}$** to be trained,
- In most cases, an ideal $f(\cdot, \mathbf{W})$ exists but is **hard to find in practice**,
- Learning is a **regression ill-posed** problem.

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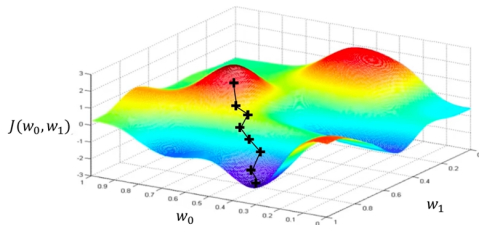
Global formalism

- **Loss:** $J(\mathbf{W}) = \sum_i \mathcal{L}(f(\mathbf{x}^{(i)}, \mathbf{W}), \mathbf{y}^{(i)})$, $i = \text{examples}$
- **Model parameters:** $\mathbf{W}^* = \text{argmin}(J(\mathbf{W}))$

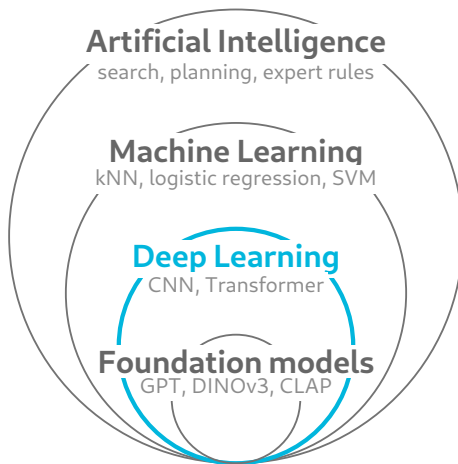
Training Algorithm

Gradient Descent:

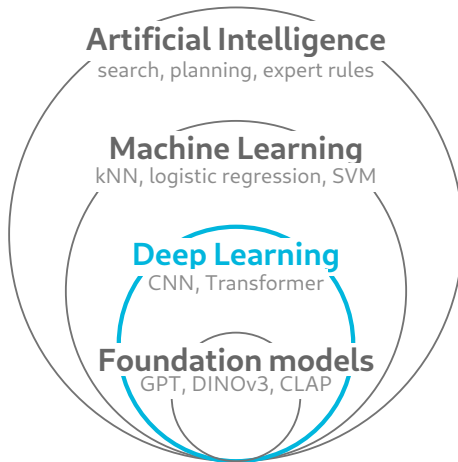
- Randomly Initialize model weights
- Compute Gradient of the Loss $\frac{\partial J(\mathbf{W})}{\partial \mathbf{W}}$
- Update weights
 $\mathbf{W} \leftarrow \mathbf{W} - \eta \frac{\partial J(\mathbf{W})}{\partial \mathbf{W}}$
- Repeat until convergence



from MIT course introtodeeplearning.com



Deep Learning



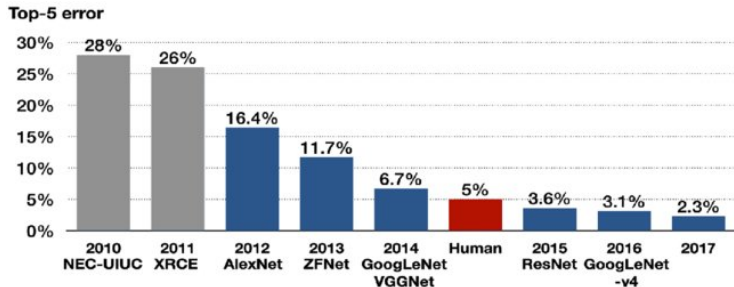
A particular case of Machine Learning

Hence, Deep Learning can be **supervised**, **unsupervised**, ...

A turning point: AlexNet (2012)

Started the reign of Deep Learning

- Using **deep** neural networks and **large** datasets
- A major breakthrough in image classification:



Source: Kang, D. Y., Duong, H. P., & Park, J. C. (2020). Application of deep learning in dentistry and implantology. Journal of implantology and applied sciences, 24(3), 148-181.

Details for the human evaluation: Russakovsky, Dieg et al.. ImageNet Large Scale Visual Recognition Challenge, <https://arxiv.org/pdf/1409.0575.pdf>

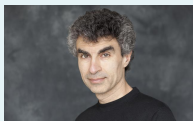
The great elders of Deep Learning (Turing Prize 2018)

Geoffrey Hinton



- Cognitive psychologist and computer scientist,
- Prof. Univ. Toronto / worked at Google until 2023,
- Known for back-propagation and Boltzmann machines (**Nobel Prize in 2024!**).

Yoshua Bengio



- Computer scientist,
- Prof. Univ. Montréal and founder of MILA,
- Known for his work on deep learning.

Yann le Cun



- Computer scientist,
- Prof. New York University / worked at META (until 2025),
- Known for his work on back-propagation and CNNs.

Where did the Deep Learning revolution come from?

- The use of GPUs for computation.
- The share of huge datasets on Internet.
- HuggingFace / Github / Arxiv: easy and open-source share of research.
- The rise of representation learning.



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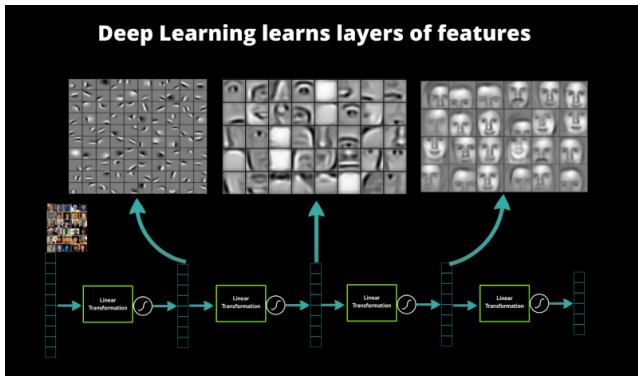
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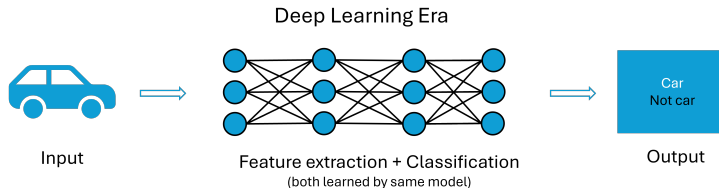
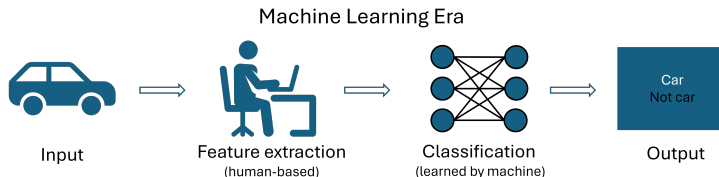
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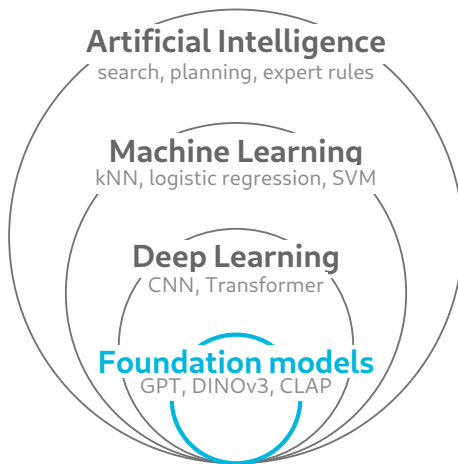
Source: <https://www.datasciencecentral.com/a-primer-on-deep-learning/>

Traditional Deep Learning (before 2020)

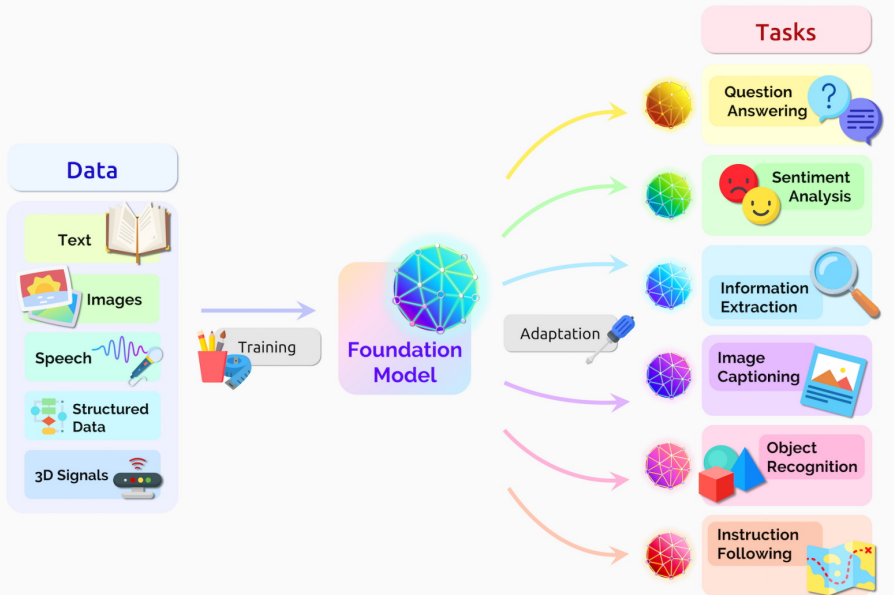


Inspired from: <https://www.softwaretestinghelp.com/data-mining-vs-machine-learning-vs-ai/>

Foundation models: a paradigm shift



Foundation models: a paradigm shift

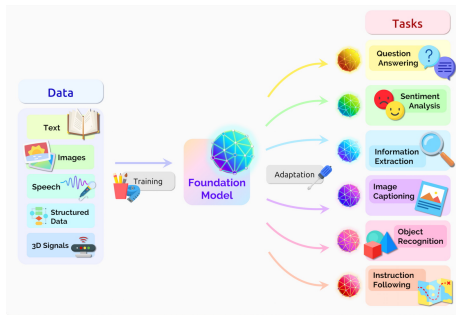


Deep Learning today - Foundation models (1/3)

Most recent Generative AI models are based on

Foundation models:

- Model trained on an Internet scale dataset
- Self-Supervised training: pretext tasks
- Generalization is not a problem anymore! All is about **particularization**



Source: <https://blogs.nvidia.com/blog/what-are-foundation-models/>

Not just Generative!

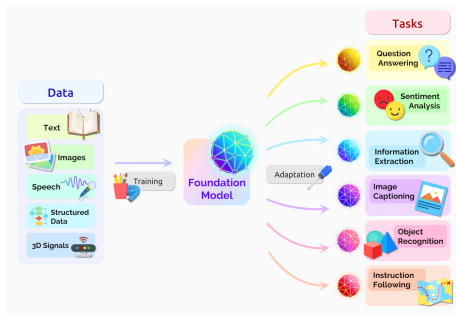
Foundation models may also be very useful for discriminative tasks, because such models understand data really well!

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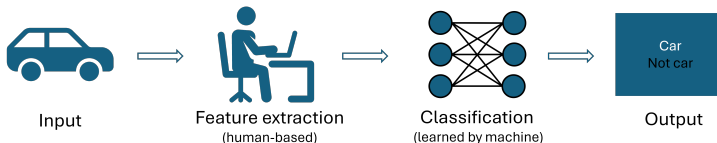
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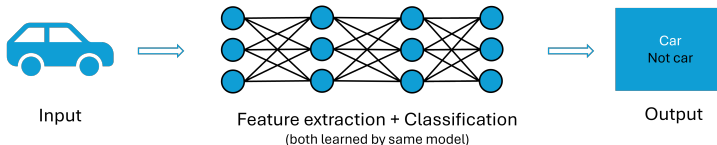
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Deep Learning today - Foundation models (2/3)

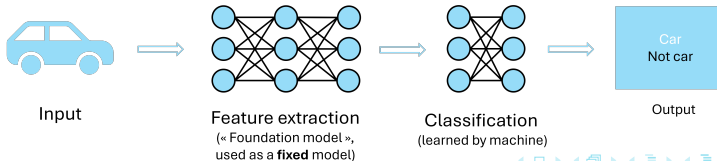
Machine Learning Era



Deep Learning Era



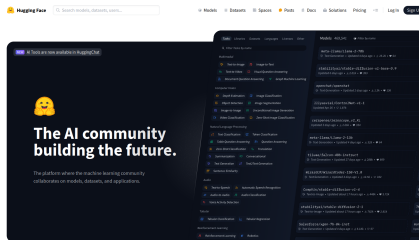
Foundation Models Era



- Foundation models are really hard to train...
 - Requires a lot (a lot!) of data
 - Requires a lot of computing power
 - Requires a lot of time
- ... But they can be used as powerful feature extractor!
 - Some open-source Foundation models exist, and are available on HuggingFace <https://huggingface.co/>
 - We will use such algorithms during our labs sessions.

Deep Learning today - Foundation models (3/3)

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Lab Session 1 (1/2) — Essential tools for data science

What you will learn from the lab:

- A dataset is a set of **points in a space** ;
- **Distances** between those points already reveal structure ;
- Sometimes, data comes broken, and you need to **clean it** first.

Lab assignments — 1h15:

- Python, IDE, virtual environments,
- Introduction to **Numpy**: arrays, shapes, broadcasting,
- Generate two clouds of points, plot them, compute the **pairwise distance matrix**, interpret it,
- Load a **real** dataset and describe it: size, missing values, annotated or not, class balance, ...

Link to the lab:

<https://mee-labs.gitlab-pages.imt-atlantique.fr/intro2ai/lab1/>

Lab Session 1 (2/2) — Your project (this afternoon)

The project is the **backbone** of this course

- **2h30 this afternoon:** form your binôme, find a dataset, define one or two tasks ;
- You will present your project **on next session**. This will be evaluated!



Your dataset must be...

- $\approx$ **1 000–10 000** samples,
- in an **easy-to-use** format,
- as **complete** as possible,
- under an **open license**.

Post your project title on Discord before you leave!

Topics are unique within a group, reserved first-come.

Discord server

- <https://discord.gg/ZuHcNMefaY> (use this link only to join the first time)
- Please immediately change your login to your first name and last name

